Edutech Solution

AI & ML INTERNSHIP

Task 7: Exploratory Data Analysis (EDA)

Tools:

•Python(Pandas, Matplotlib, Seaborn)

Dataset:

•"Automobile Dataset" or "Ecommerce Sales Data"

Hints / Mini Guide:

1. Perform Univariate Analysis (Distributions of individual variables).
2. Perform Bivariate Analysis (Correlations between variables).
3. Use Multivariate Analysis (Relationships between three or more variables).
4. Generate heatmaps for feature correlations.
5. Visualize data using Box plots to detect outliers.
6. Create Pairplots to observe group-wise data trends.
7. Summarize key insights and actionable findings.

Deliverables:

• Jupyter Notebook containing all visualizations and code
• A report highlighting the top 5 insights from the data

Final Outcome:

Internbuildsstronganalytical thinking and visual storytelling skills.

Interview Questions Related To Above Task:

• What is the importance ofEDA intheData Science lifecycle?
• Explain the difference between Univariate and Bivariate analysis.
• How do you handle skewed data discovered during EDA?
• What does a correlation coefficient of -1 indicate?
• How do you decide which visualization to use for categorical vs. numerical data?

Submission Guidelines

• **Time Window:**

You can complete the task any time between 10:00AMto10:00PM on the given day Submission l closes at 10:00 PM. [cite:

• **Self-Research Allowed:**

You are free to explore,Google,or refer to tutorials to understand concepts and complete the task effectively.

• **Debug Yourself:**

Try toresolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

• **No Paid Tools:**

Ifthetaskinvolves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

• **GitHub Submission:**

CreateanewGitHub repository for each task. Add everything used - code, datasets, screenshots, and a README.md.

• **Submit Here:**

Aftercompletingthe task, paste your GitHub repo link and submit it using the link provided.

URL : https://docs.google.com/forms/d/e/1FAIpQLSf_3RtphYvxqIgBmtDQLoiBBN6lSKrYX25WdnzXsSA38_ICYg/viewform?usp=dialog

Best of Luck